

Product Information

Product Name	GLA Rescue System
Date	March 17, 2026
Lark links	Backlog : Agile
Docs links	
Figma	Agile - Figma
Customer	A+

Introduction

This project aims to develop a web-based Employee Attendance and Leave Management System to improve the efficiency of managing employee information, attendance records, and leave requests within an organization. The system supports multiple user roles including employees, managers, HR, and administrators, each with different access permissions. It provides key functions such as attendance tracking, leave request management, employee information management, reporting, and notification services. The system is designed to reduce manual processes, improve data accuracy, and support better workforce management.

Functional Requirement

1. User Management System

- Login / Logout
 - Officer
 - Login / Logout
 - Access only their own functions
 - Manager
 - Login / Logout
 - Access only their own functions
 - Access team data under their responsibility
 - HR
 - Login / Logout
 - Access only their own functions
 - Access data of all employees
- Manage Employee Information
 - Officer

- View personal information (name, department, position, leave entitlements, attendance)
- Manager
 - View subordinate information within their department
 - View personal information (name, department, position, leave entitlements, attendance)
- HR
 - Add / Edit / View employee information for all employee
 - View personal information (name, department, position, leave entitlements, attendance)

2. Attendance System

- Daily Check-in / Check-out
 - Officer
 - Perform daily check-in / check-out
 - View confirmation status of time attendance records
 - Manager
 - View attendance status of team members
 - Perform daily check-in / check-out
 - View confirmation status of time attendance records
 - HR
 - Review and monitor attendance records of all employees
 - Perform daily check-in / check-out
 - View confirmation status of time attendance records
- Record Late Arrivals & Attendance History
 - Officer
 - View their own attendance history
 - View their own late arrival records
 - Manager
 - View their own attendance history
 - View their own late arrival records
 - View statistics of late arrivals and absences of team members
 - HR
 - View their own attendance history
 - View their own late arrival records
 - View organization-wide attendance records

3. Leave Management System

- Submit Leave Request
 - Officer
 - Submit a leave request
 - Select leave type

- View remaining leave balance before submission
 - Provide a reason for leave
 - View the status of leave requests
 - Edit the leave request before submitting.
 - Select Start date and End date leave .
- Manager
 - Submit a leave request
 - Select leave type
 - View remaining leave balance before submission
 - Provide a reason for leave
 - View the status of leave requests
 - Edit the leave request before submitting.
 - Select Start date and End date leave .
- HR
 - Submit a leave request
 - Select leave type
 - View remaining leave balance before submission
 - Provide a reason for leave
 - View the status of leave requests
 - Edit the leave request before submitting.
 - Select Start date and End date leave .
- Approve / Reject Leave
 - Manager
 - Approve / Reject team members' leave requests
 - Add comments when rejecting a leave request
 - HR
 - Approve / Reject all employee leave requests
 - Add comments when rejecting a leave request
- Track Leave Status & Notifications
 - Officer
 - View leave request status (Pending / Approved / Rejected)
 - Receive notifications of leave approval results
 - View rejection reasons when a leave request is rejected
 - Manager
 - View leave request status (Pending / Approved / Rejected)
 - Receive notifications of leave approval results
 - View rejection reasons when a leave request is rejected
 - Receive notifications when a team member submits a leave request
 - HR
 - View leave request status (Pending / Approved / Rejected)

- Receive notifications of leave approval results
- View rejection reasons when a leave request is rejected
- Receive notifications when all employee submits a leave request

4. Resource Management System

- Calendar
 - Officer
 - View personal leave date
 - View Scheduled Event Dates
 - Manager
 - View team calendar (who is on leave / who is working)
 - View Team Daily Check-In Overview
 - View personal leave date
 - View Scheduled Event Dates
 - HR
 - View all employee calendar (who is on leave / who is working)
 - View all employee Daily Check-In Overview
 - View personal leave date
 - View Scheduled Event Dates

5. Reporting System

- Attendance & Leave Reports
 - Officer
 - View personal reports (attendance / leave records)
 - Manager
 - View Department Summary Reports on Late Arrivals, Absences, and Leave
 - Export Department summary Reports on Late Arrivals, Absences, and Leave Report to Excel
 - View personal reports (attendance / leave records)
 - HR
 - Generate annual reports (attendance / leave statistics)
 - Export Organization summary Reports on Late Arrivals, Absences, and Leave Report to Excel
 - View personal reports (attendance / leave records)

6. Notification System

- System Alerts
 - Officer
 - Receive notifications for late check-ins or missed check-outs
 - Manager
 - Receive notifications for late check-ins or missed check-outs
 - Receive notifications for new leave requests from team members

- HR
 - Receive notifications for late check-ins or missed check-outs
 - Receive notifications when any employee submits a leave request

Non-Functional Requirements

- Usability
 - The system shall support both Thai and English languages for all user roles.
- Performance
 - System response time shall not exceed 2 seconds for normal operations.
 - Check-in and check-out actions shall be processed immediately.
- Security
 - The system shall enforce role-based access control (RBAC).
 - All sensitive data shall be encrypted in the database.
 - User passwords shall be stored in hashed format.
- Availability
 - System uptime shall be at least 99.5%.
- Reliability
 - The system shall maintain data consistency during concurrent access.
- Maintainability
 - The system shall allow future expansion of features.

UML

Sprint 1

Sprint Planing

- **Sprint Duration:** 1 Week
- **Sprint Goal:**

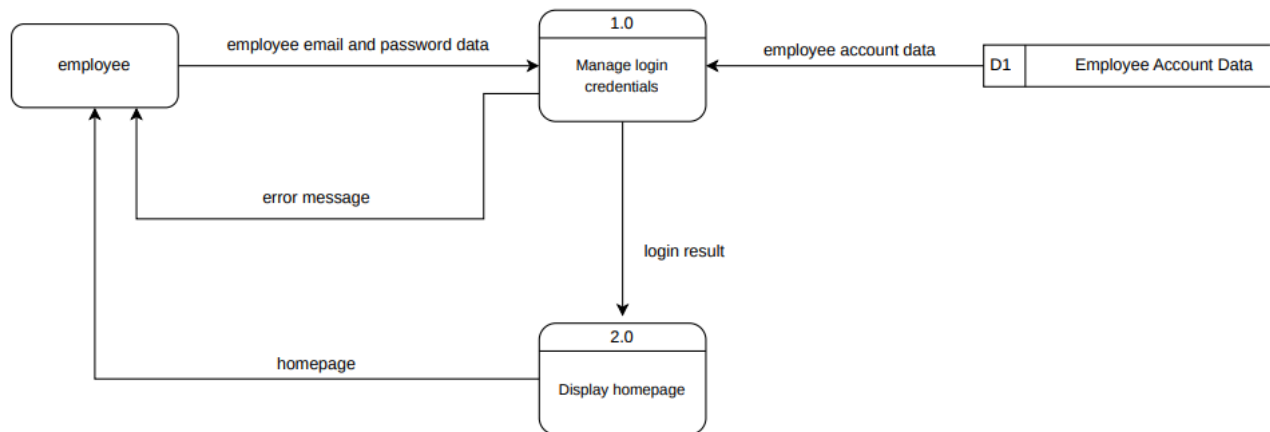
Selected Backlog Items:

ID	Backlog Item	Estimate (SP)
	Design UI login page	
	Design DFD input email/password	
	Design UI : fail login notification	

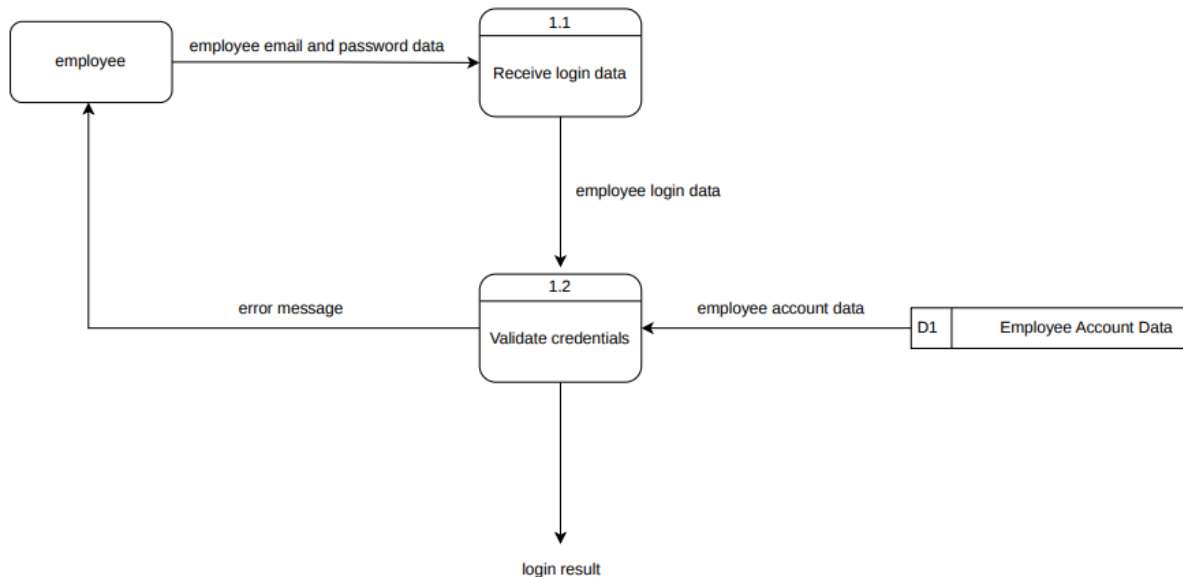
	Design UI forget password page	
	Design DFD forget password	
	Design UI HR homepage	
	Design UI worker homepage	
	Design UI management homepage	
	Design ER	

Context diagram

DFD Level 1



DFD Level 2



Process Description

Login Process

- 1.The user opens the system login page.
- 2.The user enters their email and password.
- 3.The system receives the login credentials.
- 4.The system validates the credentials against the employee account database.
- 5.If the credentials are valid, the system identifies the user's role (Officer, Manager, HR).
- 6.The system retrieves the user's profile information.
- 7.The system redirects the user to the corresponding homepage.
- 8.If the credentials are invalid, the system displays an error message and denies access.

Failed Login Handling Process

- 1.The user submits invalid credentials.
- 2.The system detects that the credentials do not match any account.
- 3.The system prevents access to the system.
- 4.The system displays an error notification indicating invalid username or password.
- 5.The user remains on the login page and may attempt to log in again.

Forgot Password Process

- 1.The user selects the “Forgot Password” option.
- 2.The system displays the password recovery form.
- 3.The user enters their registered email address.
- 4.The system verifies whether the email exists in the database.
- 5.If the email is valid, the system generates a password reset link or code.
- 6.The system sends the reset link to the user's email.
- 7.The user accesses the link and creates a new password.
- 8.The system updates the password in the database.
- 9.The user can log in using the new password.

Homepage Display Process

- 1.After successful login, the system determines the user's role.
- 2.The system retrieves relevant data for that role.
- 3.The system loads the corresponding homepage interface:
 - Officer Homepage
 - Manager Homepage
 - HR Homepage
- 4.The system displays available functions and information for the user.
- 5.The user can access system features permitted by their role.

Data dictionary - external entity description

Operator

Field	Description
Description	
Type	
Input to System	
Output from System	
Related DFD Process	

Data dictionary - data store description

Data Store	Description
Employee	Employee information
Attendance	Attendance records
Leave Request	Leave request data

Data dictionary - data flow description

In progress

ER Diagram

In progress

Test case

Test cases were created based on the user stories of the system.

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (pass/fail)
TS0001 : Verify that the system displays an error message when the user enters an incorrect username	TC01	Verify that entering incorrect username shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Click on the Username / Email field	1.Username / Email = (Incorrect) 2.Password = (Correct)	1.System processes login request 2.System displays error message: "Invalid username"	User stays on Login Page and cannot access the system	User stays on Login Page but cannot display error message	Fail

me or password				4.Enter an incorrect Username / Email 5.Click on the Password field 6.Enter a Password 7.Click the Login button		3.User remains on Login Page			
TS0001 : Verify that the system displays an error message when the user enters an incorrect username or password	TC02	Verify that entering incorrect password only shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Click on the Username / Email field 4.Enter a Username / Email 5.Click on the Password field 6.Enter an incorrect Password 7.Click	1.Username / Email = (Correct) 2.Password = (Incorrect)	1.System does not allow login 2.System displays an error message: "Invalid password" 3.User remains on the Login Page	User remains on Login Page and cannot access any system features	User stays on Login Page but cannot display error message	Fail

				the Login button					
TS0001 : Verify that the system displays an error message when the user enters an incorrect username or password	TC03	Verify that leaving username and password empty shows error	User has a valid account in the system and is on the Login Page	1.Open the web 2.System displays the Login Page 3.Leave the Username / Email field empty. 4.Click the Login button	1.Username / Email = (Empty) 2.Password = (Empty)	1.System does not allow login. 2.System displays an error message indicating that Username and Password are required. 3.User remains on the Login Page.	User stays on Login Page and cannot access the system	User stays on Login Page but cannot display error message	Fail
TS0002 : Verify that the user can log in successfully with HR credentials and is redirected to the HR homepage	TC01	Verify that the user can log in successfully with HR credentials and is redirected to the HR homepage	1. HR account already exists in the system. 2.User is on the Login Page. 3.System is running normally and connected to the database.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password	1.Username / Email = (Correct) 2.Password = (Correct) 3.Role = HR	1.Login is successful 2.System redirects the user to the HR Homepage 3. HR menu or dashboard is displayed 4.User can access	User is logged into the system and user on the HR Homepage	System allows the user to log in successfully and redirects the user to the HR Homepage. The HR dashboard is displayed correctly and the user	Pass

				rd field 6.Enter the correct password 7.Click the Login button 8.System validates the credentials 9.System identifies the user role as HR 10.System redirects the user to the HR Homepage		features that are available for the HR role only		can access the HR home page.	
TS0003 : Verify that the user can log in successfully with Worker credentials and is redirected to the Worker homepage	TC01	Verify that the user can log in successfully with Worker credentials and is redirected to the Worker homepage	1.Worker account already exists in the system. 2.User is on the Login Page. 3.System is running normally and connect	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email	1.Username / Email = (Correct) 2.Password = (Correct) 3.Role = Worker	1.Login is successful 2.System redirects the user to the Worker Homepage 3.Worker menu or dashboard is	User is logged into the system and user on the Worker Homepage	System allows the user to log in successfully and redirects the user to the Worker Homepage. The Worker dashboard is	Pass

ge			ed to the database.	5.Click on the Password field 6.Enter the correct password 7.Click the Login button 8.System validates the credentials 9.System identifies the user role as Worker 10.System redirects the user to the Worker Homepage		displayed 4.User can access features that are available for the Worker role only		displayed correctly and the user can access the Worker home page.	
TS0004 : Verify that the user can log in successfully with Manager credentials and is	TC01	Verify that the user can log in successfully with Manager credentials and is redirect	1.Manager account already exists in the system. 2.User is on the Login Page. 3.System is	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter	1.Username / Email = (Correct) 2.Password = (Correct) 3.Role = Manager	1.Login is successful 2.System redirects the user to the Manager Homepage	User is logged into the system and user on the Manager Homepage	System allows the user to log in successfully and redirects the user to the Manager Homepage	Pass

redirected to the Manager homepage		ed to the Manager homepage	running normally and connected to the database.	a valid username/email 5. Click on the Password field 6. Enter the correct password 7. Click the Login button 8. System validates the credentials 9. System identifies the user role as Manager 10. System redirects the user to the Manager Homepage		3. Manager menu or dashboard is displayed 4. User can access features that are available for the Manager role only		age. The Manager dashboard is displayed correctly and the user can access the Manager homepage.	
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Increment meets requirement in Figma (Sprint 1)

The system interface was designed using **Figma**, including:

- Login Page

- Forgot Password Page
- HR Homepage
- Worker Homepage
- Manager Homepage
- Check-in Page
- Check-out Page
- Leave Request Page

These designs represent the system prototype used during development.

Sprint 2

Sprint Planing

- **Sprint Duration:** 1 Week
- **Sprint Goal:** To design the check in, check out and leave result pages with their system flows

Selected Backlog Items:

ID	Backlog Item	Estimate (SP)
	Fixing feedback from sprint 1	
	Design UI : Check in page	
	Design UI : Check out page	
	Create DFD : Check in system (lv.?)	
	Create DFD : Check out system (lv.?)	
	Design UI : leave request page	
	Create DFD : leave request (lv.?)	
	Design UI : leave form + confirm notification	
	Create ERD for ... system	
	Write test scenario and test cases	

Context diagram

The context diagram shows the interaction between system users and the system for attendance and leave management.

Users involved in the system include:

- Officer
- Manager
- HR

These users interact with the system to perform tasks such as recording attendance, submitting leave requests, and monitoring attendance and leave information.

DFD Level 1-2

In progress

Process Description

The process descriptions explain the system workflow for the functions implemented in Sprint 2.

Check-in Process

1. The user logs into the system.
2. The user selects the Check-In function.
3. The system verifies that the user has not already checked in for the current day.
4. The system checks the current time and location (if applicable).
5. If all conditions are valid, the system records the check-in time.
6. The attendance record is stored in the database.
7. The system displays confirmation of successful check-in.

Check-out Process

1. The user logs into the system.
2. The user selects the Check-Out function.
3. The system verifies that the user has already checked in for the current day.
4. The system checks the current time and location (if applicable).
5. If conditions are valid, the system records the check-out time.
6. The attendance record is updated in the database.
7. The system displays confirmation of successful check-out.

Leave Request Process

1. The user selects the Leave Request function.
2. The system displays the leave request form.
3. The user selects the leave type.
4. The user selects the start date and end date.
5. The user enters the reason for leave.
6. The system validates the input data and checks the remaining leave balance.
7. If the request is valid, the system records the leave request in the database.
8. The leave request status is set to Pending.
9. The system sends the request to the manager or HR for approval.
10. The system displays confirmation of successful submission.

Data dictionary - external entity description

Operator

Field	Description
Description	
Type	
Input to System	
Output from System	
Related DFD Process	

Data dictionary - data store description

In progress

Data dictionary - data flow description

In progress

ER Diagram

In progress

Test case

Test cases were created to verify that the attendance and leave request functions operate correctly.

Test Scenario	Test Case ID	Test Case	Pre-Condition	Test Step	Test Data	Expect Result	Post Condition	Actual Result	Status (pass/fail)
TS0001 : Verify whether user can submit a leave request with valid inputs	TC01	Verify that user can submit leave request successfully when all required fields are filled with valid inputs	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser . 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = Medical appointment 7.Duration = 2 Days	1.System successfully saves the leave request. 2.System displays a confirmation message such as "Leave request submitted". 3.The submitted leave request appears in the Leave Request History 4.The leave details (type, date, reason, status) are displayed	The leave request is successfully recorded in the system and appears in the leave request history.	System successfully records the leave request and displays the confirmation message.	Pass

				Home page. 9.Navigate to the Leave Request Page. 10.Click the Request button. 11.Select a Leave Type. 12.Enter a valid Start Date. 13.Enter a valid End Date. 14.Enter a valid Reason for leave. 15.Enter a valid Duration for leave. 16.Click the Submit button.		correctly.			
TS0002 : Verify whether the system handles incomplete inputs on the leave	TC01	Verify system shows error when Leave Type is not selected	1.User account exists in the system. 2.User is logged into the system. 3.User	1.Open a web browser . 2.System displays the Login Page. 3.Click	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type =	1.System does not submit the leave request. 2.System displays an error	No leave request record is created.	System prevents submission and displays an error message indicating that	Pass

request page			is on the Leave Reques t Page.	on the Email / Userna me field. 4.Enter a valid userna me/ema il. 5.Click on the Passwo rd field. 6.Enter a valid passwo rd. 7.Click the Login button. 8.Syste m redirect s the user to the Employ ee Homep age. 9.Navig ate to the Leave Reques t Page. 10.Click the Reques t button. 11.Leav e the Leave Type field empty. 12.Ente r a valid	(Not selecte d) 4.Start Date = 10/03/2 026 5.End Date = 11/03/2 026 6.Reas on = Medical appoint ment 7.Durati on = 2 Days	messag e indicatin g that Leave Type is required . 3.The leave request is not saved in the system.		Leave Type is required .	
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				Start Date. 13. Enter a valid End Date. 14. Enter a valid Reason for leave. 15. Enter a valid Duration for leave. 16. Click the Submit button.					
TS0002 : Verify whether the system handles incomplete inputs on the leave request page	TC02	Verify system shows error when Start Date is not entered	1. User account exists in the system. 2. User is logged into the system. 3. User is on the Leave Request Page.	1. Open a web browser . 2. System displays the Login Page. 3. Click on the Email / Username field. 4. Enter a valid username/email. 5. Click on the Password field. 6. Enter a valid password.	1. Username / Email = (Correct) 2. Password = (Correct) 3. Leave Type = Sick Leave 4. Start Date = (Empty) 5. End Date = 11/03/2026 6. Reason = Medical appointment 7. Duration = 2 Days	1. System does not submit the leave request. 2. System displays an error message indicating that Start Date is required . 3. The leave request is not saved in the system.	No leave request record is created.	System prevents submission and displays an error message indicating that Start Date is required .	Pass

				7.Click the Login button. 8.System redirect s the user to the Employ ee Homep age. 9.Navig ate to the Leave Reques t Page. 10.Click the Reques t button. 11.Sele ct a Leave Type. 12.Leav e the Start Date field empty. 13.Ente r a valid End Date. 14.Ente r a valid Reason for leave. 15.Ente r a valid Duratio n for leave. 16.Click the					
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				Submit button.					
TS0002 : Verify whether the system handles incomplete inputs on the leave request page	TC03	Verify system shows error when End Date is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser . 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Navigate to the Leave	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = (Empty) 6.Reason = Medical appointment 7.Duration = 2 Days	1.System does not submit the leave request. 2.System displays an error message indicating that End Date is required . 3.The leave request is not saved in the system.	No leave request record is created.	System prevents submission and displays an error message indicating that End Date is required .	Pass

				Request Page. 10. Click the Request button. 11. Select a Leave Type. 12. Enter a valid Start Date. 13. Leave the End Date field empty. 14. Enter a valid Reason for leave. 15. Enter a valid Duration for leave. 16. Click the Submit button.					
TS0002 : Verify whether the system handles incomplete inputs on the leave request page	TC04	Verify system shows error when Reason is not entered	1. User account exists in the system. 2. User is logged into the system. 3. User is on the Leave Request	1. Open a web browser . 2. System displays the Login Page. 3. Click on the Email / Username	1. Username / Email = (Correct) 2. Password = (Correct) 3. Leave Type = Sick Leave 4. Start Date =	1. System does not submit the leave request. 2. System displays an error message indicating that	No leave request record is created.	System prevents submission and displays an error message indicating that Reason is required .	Pass

			t Page.	field. 4.Enter a valid userna me/ema il. 5.Click on the Passwo rd field. 6.Enter a valid passwo rd. 7.Click the Login button. 8.Syste m redirect s the user to the Employ ee Homep age. 9.Navig ate to the Leave Reques t Page. 10.Click the Reques t button. 11.Sele ct a Leave Type. 12.Ente r a valid Start Date. 13.Ente r a valid End Date.	10/03/2 026 5.End Date = 11/03/2 026 6.Reas on = (Empty) 7.Durati on = 2 Days	Reason is required . 3.The leave request is not saved in the system.			
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				14.Leave the Reason field empty. 15.Enter a valid Duration for leave. 16.Click the Submit button.					
TS0002 : Verify whether the system handles incomplete inputs on the leave request page	TC05	Verify system shows error when Duration is not entered	1.User account exists in the system. 2.User is logged into the system. 3.User is on the Leave Request Page.	1.Open a web browser . 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid password. 7.Click the Login button. 8.System	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 10/03/2026 5.End Date = 11/03/2026 6.Reason = Medical appointment 7.Duration = (Empty)	1.System does not submit the leave request. 2.System displays an error message indicating that Duration is required . 3.The leave request is not saved in the system.	No leave request record is created.	System prevents submission and displays an error message indicating that Duration is required .	Pass

				redirect s the user to the Employ ee Homep age. 9.Navig ate to the Leave Reques t Page. 10.Click the Reques t button. 11.Sele ct a Leave Type. 12.Ente r a valid Start Date. 13.Ente r a valid End Date. 14.Ente r a valid Reason for leave. 15.Leav e the Duratio n field empty. 16.Click the Submit button.					
TS0003 : Verify whether the system	TC01	Verify system saves leave request	1.User account exists in the system.	1.Open a web 2.Syste m displays	1.Usern ame / Email = (Correct)	1.Syste m success fully saves	Leave request record is stored	System success fully saves the	Pass

saves and displays the submitted leave request correctly after submission		after submission	2. User is logged into the system. 3. User is on the Leave Request Page.	the Login Page 3. Click on the Email / Username field 4. Enter a valid username/email 5. Click on the Password field 6. Enter a valid password 7. Click the Login button 8. System redirects the user to the User Homepage. 9. Click the Leave Request / Request Leave menu. 10. System displays the Leave Request Form. 11. Sele	2. Password = (Correct) 3. Leave Type = Sick Leave 4. Start Date = 15/03/2026 5. End Date = 16/03/2026 6. Reason = Flu and need rest 7. Duration = 1	the leave request. 2. System displays a confirmation message such as "Leave request submitted". 3. The submitted leave request appears in the Leave Request History 4. The leave details (type, date, reason, status) are displayed correctly.	in the system database and displayed in the user's leave request history.	leave request and displays the submitted leave request in the Leave History page correctly.	
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				ct the Leave Type. 12.Sele ct the Start Date of leave. 13.Sele ct the End Date of leave. 14.Ente r the Reason for leave in the reason field. 15.Ente r Duratio n in the field 16.Click the Submit / Send Reques t button. 17.Syst em process es and saves the leave request. 18.Navi gate to Leave History / My Leave Reques ts page to view the					
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				submitted request.					
TS0003 : Verify whether the system saves and displays the submitted leave request correctly after submission	TC02	Verify submitted leave request is displayed in leave request list	1.User account exists in the system. 2.User is logged into the system. 3.A leave request has already been submitted successfully.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the User Homepage. 9.Click the Leave Request 10.System	1.Username / Email = (Correct) 2.Password = (Correct) 3.Leave Type = Sick Leave 4.Start Date = 15/03/2026 5.End Date = 16/03/2026 6.Reason = Flu and need rest 7.Duration = 1	1.System displays the submitted leave request in the Leave Request List. 2.Leave details such as Leave Type, Start Date, End Date, Reason , Duration and Status are shown correctly. 3.Leave request status is displayed (e.g., Pending / Submitted).	Leave request remains stored in the system and visible in the leave request list.	System displays the submitted leave request in the Leave Request List with correct details.	Pass

				displays the Leave Request List page. 11. Check the leave request list for the previously submitted leave request					
TS0003 : Verify whether the system saves and displays the submitted leave request correctly after submission	TC03	Verify system displays correct leave request details	1. User account exists in the system. 2. User is logged into the system. 3. A leave request has already been submitted and saved successfully.	1. Open a web 2. System displays the Login Page 3. Click on the Email / Username field 4. Enter a valid username/email 5. Click on the Password field 6. Enter a valid password 7. Click the Login button 8. System	1. Username / Email = (Correct) 2. Password = (Correct) 3. Leave Type = Sick Leave 4. Start Date = 15/03/2026 5. End Date = 16/03/2026 6. Reason = Flu and need rest 7. Duration = 1	1. System displays the leave request details page. 2. Leave information such as Leave Type, Start Date, End Date, Reason, and Status are displayed correctly and match the submitted data. 3. No	Leave request details remain stored correctly in the system database and can be viewed by the user.	System displays the leave request details correctly and matches the submitted leave request information.	Pass

				m redirect s the user to the User Homep age. 9.Click the Leave Reques t / My Leave menu. 10.Syst em displays the Leave Reques t List page. 11.Sele ct or click the submitt ed leave request record from the list. 12.Syst em opens the Leave Reques t Details page. 13.Revi ew the leave request informat ion displaye d.		data is missing or in correc t.			
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TS0004 : Verify whether employee can check in successfully	TC01	Verify employee can check in successfully when all conditions are valid	1.Employee account exists in the system. 2.Employee is registered in the attendance system. 3.Employee is on the Home Page / Attendance Page. 4.Employee has not checked in for the current day.	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Locate the Check In button on the right panel. 10.Click the	1.Username / Email = (Correct) 2.Password = (Correct) 3.Date = Current working day 4.Check-in time = 08:55 AM	System records the employee check-in time successfully and displays the check-in status as "On time" when the employee checks in within the scheduled time.	Employee attendance record for the current day is saved in the system database.	System successfully records the check-in time and displays the check-in status as "On time" in the schedule section.	Pass
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				Check In button. 11.System records the employee's check-in time. 12.System updates the attendance status.					
TS0005 : Verify whether employee can't check in successfully	TC01	Verify system shows error when employee checks in outside allowed location	1.Employee account exists in the system. 2.Employee is registered in the attendance system. 3.System requires location verification (GPS / allowed location) for check-in. 4.Employee is currently outside	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System	1.Username / Email = (Correct) 2.Password = (Correct) 3.Current Location = Outside company area	1.System does not allow check-in. 2.System displays an error message 3.Check-in time is not recorded in the system. 4.User remains on the Employee Homepage.	No attendance record is created for the current check-in attempt.	System prevents check-in and displays an error message indicating that the employee is outside the allowed location .	Pass

			the allowed company location . 5.Employee is on the Employee Homepage. 9.Locate the Check In button on the dashboard. 10.Ensure the device location is inside the allowed company location . 11.Click the Check In button. 12.System verifies the user location .	m redirect s the user to the Employee Homepage. 9.Locate the Check In button on the dashboard. 10.Ensure the device location is inside the allowed company location . 11.Click the Check In button. 12.System verifies the user location .					
TS0005 : Verify whether employee can't check in successfully	TC02	Verify system shows error when employee checks in before	1.Employee account exists in the system. 2.Employee is registered in	1.Open a web 2.System displays the Login Page 3.Click on the	1.Username / Email = (Correct) 2.Password = (Correct) 3.Allow	1.System does not allow check-in. 2.The Check In button	No attendance record is created for this check-in attempt.	System prevents check-in before the allowed time by disabling the	Pass

		the allowed check-in time (เปลี่ยนเป็นปุ่มกดไม่ได้)	the attendance system. 3.System has a defined allowed check-in time. 4.Employee has not checked in yet for the current day. 5.Employee is on the Employee Homepage.	Email / Username field 4.Enter a valid username/email 5.Click on the Password field 6.Enter a valid password 7.Click the Login button 8.System redirects the user to the Employee Homepage. 9.Check the current time on the device. 10.Locate the Check In button on the dashboard. 11.Click the Check In button. 12.Syst	ed Check-in Time = 08:30 AM 4.Attempt Check-in Time = 07:45 AM	is disabled and cannot be clicked. 3.Check-in time is not recorded in the attendance system. 4.User remains on the Employee Homepage.		Check In button, so the employee cannot click the button.	
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				em verifies the current time against the allowed check-i n schedul e.					
TS0005 : Verify whether employ ee can't check in success fully	TC03	Verify system shows error when employ ee tries to check in again after already checkin g in (เปลี่ยน เป็นปุ่ม กดไม่ ได้)	1. Emplo yee account exists in the system. 2. Emplo yee is register ed in the attenda nce system. 3. Emplo yee has already checke d in success fully for the current day. 4. Emplo yee is on the Employ ee Homep age.	1. Open a web 2. Syste m displays the Login Page 3. Click on the Email / Userna me field 4. Enter a valid userna me/ema il 5. Click on the Passwo rd field 6. Enter a valid passwo rd 7. Click the Login button 8. Syste m redirect s the user to the Employ	1. Usern ame / Email = (Correct) 2. Pass word = (Correct) 3. First Check-i n Time = 08:45 AM 4. Secon d Attempt Time = 08:50 AM	System does not allow the employ ee to check in again after already checkin g in. The Check In button become s disable d and cannot be clicked.	Only the first check-i n record remains stored in the system for the current day.	After the employ ee checks in success fully, the Check In button is disable d and cannot be clicked again, preventi ng multiple check-i ns.	Pass

				ee Homepage. 9.Click the Check In button to perform the first check-in. 10.System records the check-in time successfully. 11.Attempt to click the Check In button again.						
TS0006 : Verify whether employee can check out successfully	TC01	Verify that the employee can check out successfully	1.Employee account exists in the system. 2.Employee has already checked in for the current day. 3.Employee is logged into the system. 4.Employee is	1.Open a web 2.System displays the Login Page 3.Click on the Email / Username field 4.Enter a valid username/email 5.Click on the Passwo	1.Username / Email = (Correct) 2.Password = (Correct) 3.Check-in Time = 08:45 AM 4.Check-out Time = 05:00 PM	1.System successfully records the check-out time. 2.System displays the recorded check-out time on the attendance or schedule panel.	Employee attendance record for the current day includes both check-in and check-out time in the system.	System records the check-out time successfully and displays it in the attendance schedule.	Pass	

			on the Employee Home page / Attendance page.	rd field 6. Enter a valid password 7. Click the Login button 8. System redirects the user to the Employee Home page. 9. Verify that the employee has already checked in for the day. 10. Locate the Check Out button on the attendance panel. 11. Click the Check Out button. 12. System records the employee's check-out time.		3. Attendance status for the day is updated.			
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TS0007 :Verify whether employ ee can't check out successfully	TC01	Verify that the system does not allow the user to check out when the user is outside the allowed location	1. Emplo yee account exists in the system. 2. Emplo yee has already checke d in for the current day. 3. Syste m requires location verificati on (GPS / allowed location) for check-o ut. 4. Emplo yee is currentl y outside the allowed compan y location . 5. Emplo yee is logged in and on the Employ ee Homep age.	1. Open a web 2. Syste m displays the Login Page 3. Click on the Email / Userna me field 4. Enter a valid userna me/ema il 5. Click on the Passwo rd field 6. Enter a valid passwo rd 7. Click the Login button 8. Syste m redirect s the user to the Employ ee Homep age. 9. Verify that the employ ee has already checke d in for the current day.	1. Usern ame / Email = (Correct) 2. Pass word = (Correct) 3. Curre nt Locatio n = Outside compan y area	1. Syste m does not allow check-o ut. 2. Syste m displays an error messag e. 3. Check -out time is not recorde d in the system.	No check-o ut record is created for this attempt.	System prevent s check-o ut becaus e the employ ee is outside the allowed location and does not record the check-o ut time.	Pass
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				10.Move the device to a location outside the allowed company area. 11.Click the Check Out button. 12.System verifies the user's location .					
TS0007 :Verify whether employee can't check out successfully	TC02	Verify that the system does not allow the user to check out without checking in first (ปุ่มกดไม่ได้)	1.Employee account exists in the system. 2.Employee has not checked in for the current day. 3.System requires employees to check in before check-out. 4.Employee is logged in and	1.Open a web browser . 2.System displays the Login Page. 3.Click on the Email / Username field. 4.Enter a valid username/email. 5.Click on the Password field. 6.Enter a valid	1.Username / Email = (Correct) 2.Password = (Correct)	1.System does not allow check-out. 2.The Check Out button is disabled or cannot be clicked. 3.Check-out time is not recorded in the system.	No check-out record is created for this attempt.	System prevents check-out because the employee has not checked in for the current day and does not record any check-out time.	Pass

			on the Employee Homepage.	password. 7.Click the Login button. 8.System redirects the user to the Employee Homepage. 9.Verify that the employee has not checked in for the current day. 10.Locate the Check Out button on the Employee Homepage. 11.Attempt to click the Check Out button.					
TS0007 :Verify whether employee can't check out	TC03	Verify that the system does not allow the user	1.Employee account exists in the system. 2.Employee	1.Open a web browser . 2.System displays	1.Username / Email = (Correct) 2.Password =	1.System does not allow another check-out	Existing check-out record remains unchanged.	System prevents duplicate check-out	Pass

success fully		to check out again after already checkin g out (ปุ่มกด ไม่ได้)	ye has already checke d in and checke d out for the current day. 3.System allows only one check-o ut record per day. 4.Empl oyee is logged in and on the Employ ee Homep age.	the Login Page. 3.Click on the Email / Userna me field. 4.Enter a valid userna me/ema il. 5.Click on the Passwo rd field. 6.Enter a valid passwo rd. 7.Click the Login button. 8.Syste m redirect s the user to the Employ ee Homep age. 9.Verify that the employ ee has already checke d in and checke d out for the current day. 10.Loca te the	(Correct)	attempt. 2.The Check Out button is disable d or unavail able. 3. No addition al check-o ut time is recorde d in the system.		attempt s becaus e the employ ee has already checke d out for the current day.	
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				Check Out button on the Employee Homepage. 11. Attempt to click the Check Out button again.					
TS0007 :Verify whether employee can't check out successfully	TC04	Verify that the system does not allow the user to check out before the allowed check-out time (ปุ่มกดไม่ได้)	1. Employee account exists in the system. 2. Employee has already checked in for the current day. 3. System has a defined allowed check-out time. 4. Current time is before the allowed check-out time. 5. Employee is logged in and on the Employ	1. Open a web browser . 2. System displays the Login Page. 3. Click on the Email / Username field. 4. Enter a valid username/email. 5. Click on the Password field. 6. Enter a valid password. 7. Click the Login button. 8. System	1. Username / Email = (Correct) 2. Password = (Correct) 3. Current Time = Before allowed check-out time	1. System does not allow check-out. 2. System displays a message indicating that check-out is not allowed yet. 3. Check-out time is not recorded in the system.	No check-out record is created for this attempt.	System prevents early check-out according to company policy and does not record the check-out time.	Pass

			ee Homep age.	m redirect s the user to the Employ ee Homep age. 9.Verify that the employ ee has already checke d in for the current day. 10.Verif y that the current time is before the allowed check-o ut time. 11.Loca te the Check Out button on the Employ ee Homep age. 12.Atte mpt to click the Check Out button.					
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Increment meets requirement in Figma (Sprint 2)

The user interface for the Sprint 2 features was designed using Figma.

The prototype includes the following pages:

- Check-in Page
- Check-out Page
- Leave Request Page

These designs demonstrate how users interact with the system to record attendance and submit leave requests.

Product Backlog

The product backlog lists the main features required for the Employee Attendance System. These backlog items are derived from the user stories and prioritized based on their importance to the system. The development work is planned across two sprints. Sprint 1 focuses on system security and user access, while Sprint 2 focuses on attendance management and leave request functionality.

Product Backlog Item

ID	Backlog Item	Type	Priority	Estimate (SP)	Planned Sprint
PB1	Secure Login System	Feature	High	5	Sprint 1
PB2	Role-Based Access Control	Feature	High	5	Sprint 1
PB3	Forgot Password Function	Feature	Medium	3	Sprint 1
PB4	Role-Based Homepage Dashboard	Feature	High	3	Sprint 1
PB5	User Profile Viewing	Feature	Medium	2	Sprint 1
PB6	Check-in Function	Feature	High	3	Sprint 2

PB7	Check-out Function	Feature	High	3	Sprint 2
PB8	Attendance History Viewing	Feature	Medium	3	Sprint 2
PB9	Leave Request Submission	Feature	High	5	Sprint 2
PB10	Leave Approval by Manager/HR	Feature	High	5	Sprint 2
PB11	Leave Status Tracking	Feature	Medium	3	Sprint 2
PB12	Notification System	Feature	Medium	3	Sprint 2
PB13	Attendance & Leave Reports	Feature	Low	5	Sprint 2

User Story

Sprint 1

As a system user, I want to log in securely and access features based on my role, so that data is protected and I can use only authorized functions.

Sprint 2

As an employee, I want to check in and check out daily so that my attendance is recorded and I want to submit a leave request so that I can request time off.

As an Employee, I want to submit a leave request so that I can request time off.

Persona

Photo			
Name	Ploy Supaporn Wongchai	Chai Somchai Rattanakul	Nat Nattapon Srisawat
Age	32	42	27
Role	HR Manager	Department Manager	Employee (Officer)
Occupation	HR Manager	Marketing Department Manager	Marketing Officer
Personality	Detail-oriented, systematic, multitasker	Busy, decisive, meeting-driven	Tech-savvy, efficiency-focused, punctual
Tech Savviness	Moderate – High	Moderate	High
Problems	Paper-based forms get lost, manual leave calculation, no real-time attendance visibility, data scattered in Excel	Unaware of team absences until last minute, misses approval requests during meetings, hard to plan projects	Unsure if request was received, long wait for approval, forgets to check-in/out
Expectations	Dashboard with pending requests, auto leave balance, one-click reports, real-time team calendar	Clear notifications, team calendar view, simple approval process, daily staffing count	Prominent check-in/out button, leave balance on dashboard, status tracking, check-out reminder
Key Features Used	Leave approval, employee management, reports, leave policy	Leave approval, team calendar, attendance reports, notifications	Check-in/out, leave request, leave balance, attendance history

	settings, notifications		
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User Requirements & User Stories

No.	Persona	User Requirement	User Story	Acceptance Criteria
1	System User	UR-01: User Authentication & RBAC	As a system user, I want to log in securely and access features based on my role, so that data is protected and I can use only authorized functions.	- Login with username and password - Role-based access: Officer accesses personal functions only - Role-based access: Manager accesses team data + personal functions - Role-based access: HR accesses all employee data + management functions - Role-based access: Admin accesses full system configuration - Session timeout and secure logout - Failed login lockout after repeated attempts
2	HR Officer	UR-02: Employee Information Management	As an HR officer, I want to manage employee information in one centralized system, so that employee records are accurate and up to date.	- HR can Add / Edit / View all employee records - Manager can view personal info and subordinate info within their department - Officer can view their own personal info only (name, dept, position, leave entitlements, attendance)

3	Employee	UR-03: Attendance Management System	As an employee, I want to check in and check out daily so that my attendance is recorded and I want to submit a leave request so that I can request time off.	- All roles can perform daily check-in / check-out with timestamp - All roles can view own check-in/check-out confirmation status - All roles can view own attendance history and late arrival records - Manager can view attendance status and late arrival statistics of team members - HR can review and monitor attendance records of all employees - System flags late arrivals automatically
4	Employee	UR-04: Leave Management System	As an employee, I want to check in and check out daily so that my attendance is recorded and I want to submit a leave request so that I can request time off.	- All roles can submit leave request (select leave type, select start/end date, provide reason) - All roles can view remaining leave balance before submission - All roles can edit leave request before final submission - All roles can view status of leave requests (Pending / Approved / Rejected) - All roles can view rejection reason when request is rejected - All roles receive notification of leave approval results - Manager can Approve / Reject team members' leave requests with optional comment

				- HR can Approve / Reject all employee leave requests with optional comment - Manager and HR receive notification when a leave request is submitted
5	Manager	UR-05: Resource & Workforce Planning	As a manager, I want to view team availability on a calendar, so that I can plan workforce resources effectively.	- Officer can view personal leave dates and scheduled event dates on calendar - Manager can view team calendar (who is on leave / who is working) - Manager can view team daily check-in overview - HR can view all employee calendar and daily check-in overview
6	HR Officer	UR-06: Reporting & Analytics	As an HR officer, I want to generate attendance and leave reports, so that management can make data-driven decisions.	- Officer can view personal attendance and leave reports - Manager can view department summary reports (late arrivals, absences, leave) - Manager can export department summary reports to Excel - HR can generate annual attendance and leave statistics reports - HR can export organization-wide summary reports to Excel

7	System User	UR-07: Notification & Alert System	As a system user, I want to receive important notifications and alerts, so that I can take action promptly.	- All roles receive notification for late check-ins or missed check-outs - Manager receives notification when a team member submits a leave request - HR receives notification when any employee submits a leave request - Notifications delivered in-app and / or via email
8	User	UR-08: System Quality, Performance & Localization	As a user, I want the system to be fast, reliable, and available in Thai and English, so that I can use it efficiently anytime.	- System response time does not exceed 2 seconds for normal operations - Check-in and check-out actions are processed immediately - System uptime is at least 99.5% - Supports Thai and English languages (toggle for all roles) - Role-based access control (RBAC) enforced throughout - All sensitive data encrypted in the database - User passwords stored in hashed format - System maintains data consistency during concurrent access - System allows future expansion of features

WOW Factor - Sprint 1

Sprint 1 focused on building the foundation of the (ชื่อเว็บ) — secure login and role-based access control. While authentication may seem like a standard feature, our implementation goes beyond a simple username/password check by delivering a fully personalized and context-aware experience for every user.

Smart Role-Based Homepage Redirect

- **What it does** : After logging in, the system automatically identifies the user's role and redirects them to a customized homepage designed specifically for that role — HR, Manager, or Officer — without any extra steps
- **How it works** : Upon successful login, the system retrieves the user's role from the database and dynamically loads the corresponding interface. Each homepage displays only the features, menus, and data relevant to that role. An HR user sees employee management tools and pending approvals; a Manager sees their team's attendance overview; an Officer sees their own check-in status and leave balance
- **Why it stands out** : Most basic systems show the same interface to all users and rely on permission flags to hide irrelevant menus. Our system takes a different approach — each role gets a fully tailored dashboard that reduces visual clutter, improves usability, and ensures users can act immediately upon login without navigating through unrelated features.

Intelligent Failed Login Handling

- **What it does** : The system handles login failures gracefully by providing clear, specific error
- **How it works** : When a user enters incorrect credentials, the system detects whether the username, password, or both are invalid and displays an appropriate error message. If the username does not exist, an 'Invalid username' message is shown. If the password is wrong, an 'Invalid password' message appears. If both fields are left empty, the system prompts the user to fill in all required fields. The user remains on the login page and can retry immediately.
- **Why it stands out** : Rather than showing a generic 'Login failed' message for all error types — which is common in basic systems — our implementation gives users actionable feedback so they can correct their input quickly. This reduces user frustration and support requests while maintaining system security.

WOW Factor - Sprint 2

Sprint 2 introduced the core daily-use features of the system — attendance check-in/check-out and leave request management. The WOW factors in this sprint focus on

smart UI controls and real-time feedback that make the system intuitive and error-resistant for everyday use.

Context-Aware Check-In / Check-Out Button States

- **What it does** : The Check-In and Check-Out buttons are not static — they dynamically enable or disable themselves based on the employee's current context, preventing invalid actions before they happen.
- **How it works** : The system evaluates three conditions in real time: (1) whether the current time falls within the allowed check-in or check-out window, (2) whether the employee has already checked in or checked out for the day, and (3) whether the employee's location is within the allowed company area. If any condition is not met, the button is automatically disabled and cannot be clicked. For example, if an employee has already checked in, the Check-In button becomes disabled immediately — and the Check-Out button remains disabled until the allowed check-out time.
- **Why it stands out** : A typical attendance system relies on server-side validation after the user submits an action, meaning users can attempt invalid check-ins and only receive an error after the fact. Our approach disables the button proactively — the UI itself communicates what is and isn't allowed at any given moment. This eliminates accidental duplicate entries, reduces error messages, and creates a much smoother daily experience for employees.

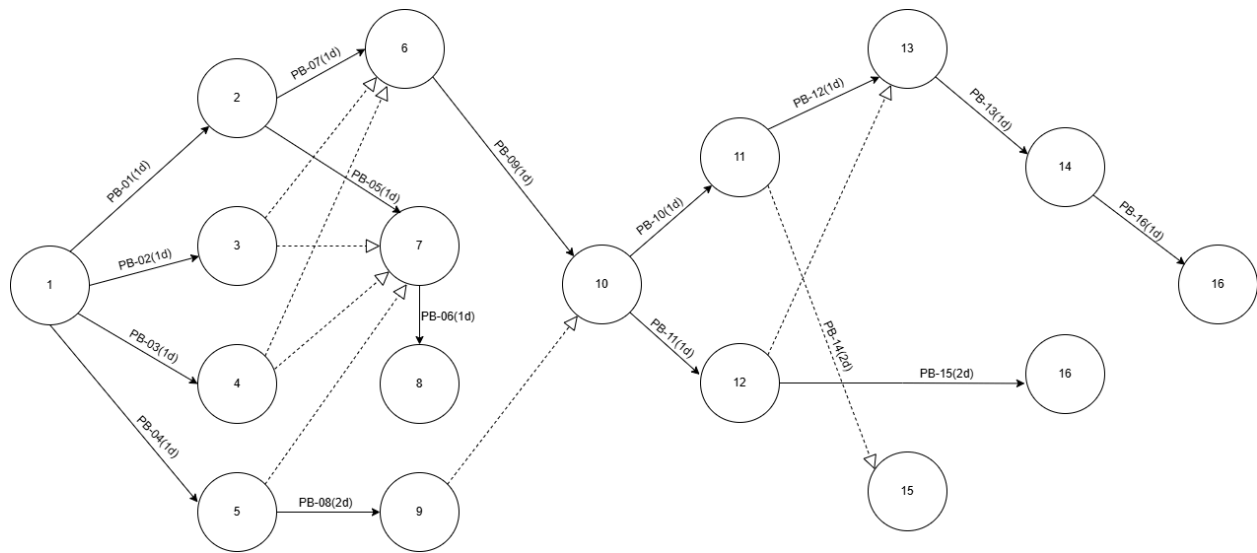
Real-Time Leave Balance Display Before Submission

- **What it does** : When an employee opens the leave request form, their remaining leave balance is displayed immediately — before they fill in any details — so they can make informed decisions without navigating away.
- **How it works** : The system retrieves the employee's leave entitlement and deducts any approved or pending leave days to calculate the current available balance. This balance is shown prominently on the leave request page broken down by leave type (e.g., Sick Leave, Annual Leave, Personal Leave). As the employee selects leave type and enters the duration, the system updates the display dynamically to show how many days would remain after the request.
- **Why it stands out** : In many systems, employees submit leave requests and only discover they have insufficient balance after the request is rejected. This creates unnecessary back-and-forth between employees, managers, and HR. By surfacing the leave balance at the point of decision — and updating it dynamically as the form is filled — our system empowers employees to self-manage their leave without surprises, and reduces the volume of invalid submissions that HR and managers have to handle.

PERT Diagram

In progress

ID	Activity	Duration (days)	Predecessor
PB-01	Design UI login page	1	-
PB-02	Design UI fail login notification	1	-
PB-03	Design UI forget password page	1	-
PB-04	Design UI homepages (HR/Worker/Manager)	1	-
PB-05	Design DFD (login system)	1	PB-01, PB-02, PB-03, PB-04
PB-06	Design ER	1	PB-05
PB-07	Test cases (login, fail, forgot password)	1	PB-01, PB-02, PB-03
PB-08	Test cases (homepages HR/Worker/Manager)	2	PB-04
PB-09	Improve UI (feedback Sprint 1)	1	PB-07, PB-08
PB-10	Design UI check-in & check-out	1	PB-09
PB-11	Design UI leave request & form	1	PB-09
PB-12	Design DFD (check-in/out & leave request)	1	PB-10, PB-11
PB-13	Design ER (Sprint 2)	1	PB-12
PB-14	Test cases (check-in & check-out)	2	PB-10
PB-15	Test cases (leave request)	2	PB-11
PB-16	Bug fix UI (check-in/out)	1	PB-14



Critical Path Analysis

Activity	Dur	ES	EF	LS	LF	TF
PB-01	1	0	1	1	2	1
PB-02	1	0	1	1	2	1
PB-03	1	0	1	1	2	1
PB-04	1	0	1	0	1	0
PB-05	1	1	2	6	7	5
PB-06	1	2	3	7	8	5
PB-07	1	1	2	2	3	1
PB-08	2	1	3	1	3	0
PB-09	1	3	4	3	4	0
PB-10	1	4	5	4	5	0
PB-11	1	4	5	5	6	1
PB-12	1	5	6	6	7	1
PB-13	1	6	7	7	8	1

PB-14	2	5	7	5	7	0
PB-15	2	5	7	6	8	1
PB-16	1	7	8	7	8	0